DILLAN IMANS

Medical AI Researcher

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SUMMARY

Medical AI researcher and M.Sc. student in AI in Biomedicine at TU Munich (Elite Network of Bavaria). First-author papers at a MICCAI workshop, IEEE BIBM and Diagnostics; co-author on an accepted npj Digital Surgery paper. Research focus: cross-modal learning, domain adaptation, knowledge distillation, JEPA, and world models; applied to medical imaging and surgical robotics.

EDUCATION

M.Sc. AI in Biomedicine, Technical University of Munich **Oct 2026 – present**
Member of Elite Network of Bavaria, Imaging and Sensing focus

B.S. Computer Science & Engineering, Sungkyunkwan University **Aug 2022 - Jun 2026**
95.7 / 100 · Dean's List · Academic Excellence Scholarship

PUBLICATIONS

Full list: scholar.google.com/citations?user=o-TtUq0AAAAJ

First author

1. **Imans, D.**, Bui, P.N., Le, D.T., & Choo, H. "Clinical Graph-Mediated Distillation for Unpaired MRI-to-CFI Hypertension Prediction." *EMA4MICCAI 2026*.
2. **Imans, D.**, Bui, P.N., Le, D.T., & Choo, H. "Unsupervised Domain Adaptation with SAM-RefiSeR for Enhanced Brain Tumor Segmentation." *IEEE BIBM 2025*.
3. ***Imans, D.**, Abuhmed, T., Alharbi, M., & El-Sappagh, S. "Explainable Multi-Layer Dynamic Ensemble Framework Optimized for Depression Detection and Severity Assessment." Diagnostics, 14(21), 2385 (2024)*.
4. **Imans, D.**, Bui, P.N., Le, D.T., & Choo, H. "Shared Semantic Codebook Distillation for Unpaired Cross-Modal Medical Classification." *Preprint*.

Co-authored

5. Xu, M., Shen, D., Zhang, J., Yip, H.C., Gao, Y., Chen, C., **Imans, D.**, et al. "Generalized Recognition of Basic Surgical Actions Enables Skill Assessment and Vision-Language-Model-based Surgical Planning." *Accepted, npj Digital Surgery*.
6. He, Y., Guo, P., Xu, M., Li, Z., **Imans, D.**, et al. "Cosmos-H-Surgical: Learning Surgical Robot Policies from Videos via World Modeling." *Preprint*.
7. Xu, M., Huang, Z., **Imans, D.**, Ye, Y., Zhang, X., & Dou, Q. "SAP-Bench: Benchmarking Multimodal Large Language Models in Surgical Action Planning." *Preprint*.

EXPERIENCE

Graduate Medical AI Researcher **Oct 2026 - Present**
CAMP, TUM, Germany

- Developing a self-supervised multimodal OR encoder with a graph JEPA architecture and higher-order graph representations, fusing multi-view video, 3D point clouds, audio and kinematic telemetry for surgical scene understanding and scene graph generation

Undergraduate Medical AI Researcher & Engineer **Oct 2024 - Jun 2026**
Superintelligence Lab, SKKU, South Korea

- Developed a two-phase unsupervised domain adaptation framework (frequency alignment + SAM-guided refinement) for cross-modality brain tumor segmentation; +9.0 Dice over strongest baseline across 6 BraTS 2021 settings [2]
- Built a graph-mediated distillation framework for unpaired MRI-to-fundus hypertension prediction across disjoint cohorts; +5.2 AUC, +3.2 AUPRC over best baseline, MRI-free at inference [1]
- Built a codebook-based distillation framework for unpaired cross-modal classification; +5.7 and +2.5 macro-F1 on OCT→fundus and CT→X-ray [4]
- Built an end-to-end retinal screening system (optic disc/cup segmentation, vessel analysis) and its backend; ONNX-optimized for low-latency inference and delivered to a clinical partner

Undergraduate Medical AI Researcher **Jun 2024 - Jun 2026**
Labren, CUHK, Hong Kong

- Curated a dataset and ran initial model experiments for a multimodal surgical robotics paper on learning robot policies via world modeling, in collaboration with NVIDIA [6]
- Built preprocessing infrastructure and an annotation QA pipeline for the first large-scale surgical action recognition dataset (11,915 clips) [5]
- Curated a dataset for multimodal LLM benchmarking for surgical action planning [7]

Undergraduate Medical AI Researcher **Feb 2024 - Oct 2024**
Infolab, SKKU, South Korea

- Built an explainable multi-layer dynamic ensemble for depression detection and severity assessment; 88.3% accuracy across 2,763 clinical records with SHAP interpretability, benchmarked against 20+ algorithms [3]

SERVICE

Reviewer, EMA4MICCAI 2026 (MICCAI workshop)

TECHNICAL SKILLS

Research areas: Cross-modal learning · Domain adaptation · Knowledge distillation · Medical image segmentation · Surgical scene understanding · Multimodal learning
Programming Languages: Python · C/C++ · Java · TypeScript
ML: PyTorch · NumPy · Pandas · scikit-learn · OpenCV · HuggingFace
Medical Imaging: MONAI · nnU-Net · SimpleITK · NiBabel · pydicom
Tooling & Infra: Git · GitHub Actions · Linux · Docker · Bash · HuggingFace · wandb · TensorBoard
Web: Next.js · Tailwind · FastAPI · PostgreSQL · Supabase

PROJECTS

Vigilia Mundi — *vigiliamundi.com*
Live geospatial dashboard aggregating global disaster and epidemic data (GDACS, WHO); Next.js, FastAPI, PostgreSQL.

LANGUAGES

English, Indonesian (Fluent) · Mandarin, Korean (Conversational) · German (A2, working towards B2)